Yan-Shuo Li

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Research Interests

Safe and adaptive decision-making; planning under uncertainty and POMDPs; safety guarantees and uncertainty quantification; multi-robot planning and search.

Education

Virginia Tech Aug. 2026 – Present
PhD Student in Aerospace Engineering *Blacksburg, VA*

- Advisor: Prof. Qi Heng Ho

National Central University (NCU) Sept. 2022 – June 2024
MS in Mathematics *Taoyuan, Taiwan*

- Thesis: Multi-Robot Search in a 3D Environment Using Submodularity With Matroid Intersection Constraints [PDF]
- Thesis Advisor: Prof. Kuo-Shih Tseng

Ming Chuan University (MCU) Sept. 2015 – June 2019
BS in Computer and Communication Engineering *Taoyuan, Taiwan*

- Undergraduate thesis: Detection of Atrial Fibrillation Using 1D Convolutional Neural Network
- Project Advisor: Prof. Chaur-Heh Hsieh

Research Experience

VT Aerospace Engineering Aug. 2026 – Present
Graduate Research Assistant with Prof. Qi Heng Ho *Blacksburg, VA*

- Exploring uncertainty-quantification and safety-assurance methods for reliable decision-making under distribution shifts and unexpected failures.

NCU Robotic Search Lab [Website] Aug. 2024 – July 2026
Research Assistant with Prof. Kuo-Shih Tseng *Taoyuan, Taiwan*

- Led the Robot Chef project, designing an intelligent robotic cooking system using Vision-Language-Action (VLA) models and submodularity-based methods, in partnership with Yo-Kai Express, a U.S. start-up.
- Developed a multi-robot exploration algorithm (HOPPER) and submitted one first-author research paper.

NCU Robotic Search Lab July 2022 – July 2024
Graduate Research Assistant with Prof. Kuo-Shih Tseng *Taoyuan, Taiwan*

- Developed theoretically grounded algorithms for multi-robot search using submodular optimization and matroid constraints.
- Evaluated the algorithms through large-scale simulations and real-world UAV experiments.
- Published three first-author publications in ICRA and IJRR.

MCU Interactive Media Lab Sept. 2017 – June 2018
Undergraduate Research Assistant with Prof. Chaur-Heh Hsieh *Taoyuan, Taiwan*

- Developed a 1D CNN for atrial fibrillation detection and published a research paper in MDPI Sensors.

Publications

Journal Articles

1. **Multi-robot search in 3d environments using submodularity with matroid intersection constraints**
   Yan-Shuo Li, Kuo-Shih Tseng
   The International Journal of Robotics Research (IJRR), 2025
   [Video] [PDF]
2. **Detection of atrial fibrillation using 1d convolutional neural network**
   Chaur-Heh Hsieh, Yan-Shuo Li, Bor-Jiunn Hwang, Ching-Hua Hsiao
   Sensors, 2020
   [PDF]

Conference Proceedings

1. **Computation-aware multi-object search in 3d space using submodular tree**
   Yan-Shuo Li, Kuo-Shih Tseng
   IEEE International Conference on Robotics and Automation (ICRA), 2024
   [Video] [PDF]
2. **Multi-robot search in a 3d environment with intersection system constraints**
   Yan-Shuo Li, Kuo-Shih Tseng
   IEEE International Conference on Robotics and Automation (ICRA), 2024
   [Video] [PDF]

Submitted Manuscript

1. **HOPPER: Multi-Robot Exploration through Hexagonal Coverage Path Planning**
   Under Review
   Yan-Shuo Li, Kuo-Shih Tseng

Invited Talks

1. **UAV Search and Map Exploration via Submodularity**
   NCU Math & AI Workshop, August 6, 2025.
   [Website]

Patents

1. **Multi-Vehicle Spatial Balanced Coverage System and Method**
   Kuo-Shih Tseng, Yan-Shuo Li, Pao-Te Lin
   US Patent, US20260038371A1, 2026 | *R.O.C. Patent*, I923020, 2026

Grants and Research Funding

1. **Digital Twins Technology of Intelligent Chefs**
   Sponsored by Yo-Kai Express Inc. (Aug. 2025 – July 2026)
2. **Optimality, Invariance, and Adaptation of Multi-robot Informative Path Planning**
   Sponsored by National Science and Technology Council, Taiwan (Aug. 2024 – July 2025)
3. **Submodular Tree for Informative Path Planning**
   Sponsored by National Science and Technology Council, Taiwan (Aug. 2023 – July 2024)
4. **Inverse Reinforcement Learning for Multi-robot Informative Path Planning**
   Sponsored by National Science and Technology Council, Taiwan (Sept. 2022 – July 2023)

Work Experience

TAO Info [**Website**] Feb. 2021 – June 2022
AI Engineer *Taipei, Taiwan*

- Designed a pharmaceutical inspection system, reducing pharmacists' medication inspection time by an average of 30% and decreasing the number of pharmacists needed by 50%.
- Developed a medication classification model using contrastive learning methods, achieving a detection error rate of 0.5% during on-site hospital testing.
- Developed multi-CNN models for inspecting damage in LCD displays, achieving 92% accuracy and reducing inspection time on the production line by about 30%.

EverComm Singapore (Taiwan Branch) [**Website**] Mar. 2020 – Jan. 2021
Software Engineer *New Taipei, Taiwan*

- Developed a real-time solar-panel monitoring platform that aggregates multi-sensor data into MySQL and serves it via RESTful APIs to front-end clients.
- Streamlined back-end processing of raw sensor data into readable formats, reducing processing time by 75% with NumPy.
- Developed an autoencoder-based anomaly detector for cooling towers and associated equipment, automatically notifying managers of detected irregularities.

National Chung-Shan Institute of Science and Technology (NCSIST) [**Website**] Jan. 2019 – June 2019
Machine Learning Intern *Taoyuan, Taiwan*

- Developed a flood forecasting system by analyzing historical time-series data on river levels and rainfall, utilizing deep learning models for enhanced prediction accuracy.
- Fine-tuned a CNN-LSTM model for a flood forecasting project, achieving a 14% reduction in root-mean-square deviation.
- Designed and implemented a face recognition access control system at NCSIST, leveraging the FaceNet model for accurate and efficient identity verification.

Extracurricular Activities

Cedar Point Amusement Park June 2017 – Sept. 2017
Park Services Associate *Sandusky, OH, United States*

- Assisted approximately 20 customers daily by providing directions and addressing park-related inquiries to enhance their experience.
- Maintained the cleanliness of tables, pavilions, and food patios throughout the park.
- Worked 40 hours per week.

MCU People to People International Student Chapter [**Website**] July 2016 – June 2017
Club Leader *Taoyuan, Taiwan*

- Led the club with over 20 international students.
- Organized club schedule including meetings, activities, and events.
- Won the "Best Club Award" (1 out of 7 clubs).

Skills

Programming Languages: Python, C/C++, Java, JavaScript, SQL

Software: PyTorch, TensorFlow, OpenCV, Git, Docker, ROS, Gazebo, NVIDIA Isaac Sim

Languages: Chinese (Native), English (Fluent)